EDUGENIE

AI-Powered Personalized Learning Assistant

PROJECT PLANNING PHASE

Project Documentation

Item	Details
Project Title	EduGenie – AI-Powered Personalized Learning Assistant
Project Type	Academic / College Project
Domain	Artificial Intelligence & Education
Technology	Generative AI, Web Technologies, Database
Development Model	Agile
Document Phase	Project Planning Phase
Academic Year	2026–2027

1. Introduction

EduGenie is an AI-powered personalized learning assistant designed to help students learn academic concepts in an interactive and personalized manner. The system uses Artificial Intelligence and Generative AI techniques to provide explanations, answer questions, generate learning materials, create quizzes, and assist students in revising their subjects.

The purpose of this planning phase is to define the project's objectives, scope, requirements, work distribution, development schedule, milestones, risks, and expected deliverables before implementation begins.

2. Project Objectives

The major objectives of EduGenie are:

To develop an AI-based learning assistant for students.

To provide personalized explanations based on the learner's requirements.

To allow students to ask academic questions using natural language.

To generate quizzes and practice questions automatically.

To provide concise summaries and study materials.

To help students revise topics efficiently.

To provide an easy-to-use and responsive learning interface.

To reduce the time required to search for basic academic explanations.

3. Project Scope

3.1 In-Scope

Module	Description
User Management	Student registration and login
AI Chat Assistant	Students can ask questions and receive AI-generated responses
Topic Explanation	Provides explanations for selected academic topics
Summarization	Generates short summaries of learning content
Quiz Generation	Creates questions based on a selected topic
Learning Materials	Provides organized study content
Progress Tracking	Records basic learning and quiz progress
Dashboard	Displays learning activities and available features

3.2 Out-of-Scope

Item	Description
Official Certification	EduGenie does not issue academic certificates
Professional Tutoring	The system is not a replacement for qualified teachers
Examination Evaluation	It does not officially evaluate university examinations
Offline AI Model Training	Training a large language model from scratch is outside the project scope

4. Stakeholders

Stakeholder	Responsibility / Interest
Students	Primary users of the system
Project Team	Design, develop and test the application
Project Guide	Provides academic guidance and reviews progress
Department	Provides project requirements and evaluation
Administrator	Manages application-level information
Faculty	Potential users for educational content and feedback

5. Functional Requirements

ID	Requirement
FR01	The system shall allow users to register and log in.
FR02	The system shall provide an AI-based question-answering interface.
FR03	The system shall generate explanations for academic topics.
FR04	The system shall generate summaries from learning content.
FR05	The system shall generate quizzes based on selected topics.
FR06	The system shall display quiz results.
FR07	The system shall maintain basic user learning information.
FR08	The system shall provide a personalized dashboard.
FR09	The system shall allow users to navigate between learning modules.
FR10	The system shall provide appropriate error messages when required.

6. Non-Functional Requirements

Category	Requirement
Performance	The application should provide responses within a reasonable time.
Usability	The interface should be simple and student-friendly.
Security	User information should be protected from unauthorized access.
Reliability	The application should handle common user operations consistently.
Scalability	The system architecture should allow additional features in the future.
Maintainability	The source code should be modular and easy to maintain.
Compatibility	The application should work on commonly used modern browsers.
Availability	The application should be accessible whenever the required services are available.

7. Proposed System Modules

Module 1 – User Authentication

Handles registration, login, logout, and basic user information.

Module 2 – AI Learning Assistant

Provides a conversational interface through which students can ask questions and obtain AI-generated explanations.

Module 3 – Personalized Learning

Uses the student's selected topics and learning activities to provide more relevant learning assistance.

Module 4 – Quiz Generation

Generates practice questions from selected subjects or topics and evaluates the submitted answers.

Module 5 – Summary Generator

Converts lengthy learning content into concise study notes.

Module 6 – Progress Dashboard

Displays learning activities, quiz performance, and other relevant progress information.

8. Technology Stack

Layer	Proposed Technology
Frontend	HTML, CSS, JavaScript / React
Backend	Python / Node.js
AI Integration	Generative AI API
Database	Firebase / MongoDB / MySQL
Authentication	Firebase Authentication / Custom Authentication
Version Control	Git & GitHub
Development Environment	VS Code
Hosting	Cloud-based deployment platform

The exact technologies may be modified according to the team's implementation.

9. Development Methodology

The project will follow an Agile development methodology.

The application will be divided into smaller features and developed through short iterations called sprints. At the end of each sprint, completed features will be tested and reviewed.

Agile Development Cycle

Planning → Development → Testing → Review → Improvement → Next Sprint

This approach allows the project team to identify problems early and incorporate improvements throughout development.

10. Work Breakdown Structure (WBS)

WBS ID	Task	Major Activities
1.0	Project Initiation	Problem identification, objectives, feasibility
2.0	Requirement Analysis	Functional and non-functional requirements
3.0	System Design	Architecture, database and UI design
4.0	AI Module	AI integration and prompt design
5.0	Frontend Development	Pages, dashboard and user interface
6.0	Backend Development	APIs, authentication and business logic
7.0	Database	Database design and integration
8.0	Testing	Unit, integration and system testing
9.0	Deployment	Hosting and configuration
10.0	Documentation	Reports, screenshots and final documentation

11. Sprint Planning

Sprint	Duration	Main Activities	Expected Output
Sprint 1	Week 1–2	Requirement analysis and project setup	Requirements document
Sprint 2	Week 3–4	UI/UX and system architecture	Design prototype
Sprint 3	Week 5–6	Authentication and database	Working user module
Sprint 4	Week 7–8	AI assistant integration	Working AI chatbot
Sprint 5	Week 9–10	Quiz and summary modules	Learning features
Sprint 6	Week 11–12	Dashboard and progress tracking	Integrated application
Sprint 7	Week 13–14	Testing and debugging	Tested application

Sprint 8 Week 15–16 Deployment and documentation Final project

12. User Stories

ID User Story Priority

US01 As a student, I want to create an account so that I can access the learning system. High

US02 As a student, I want to log in securely so that I can access my learning information. High

US03 As a student, I want to ask the AI questions so that I can understand difficult concepts. High

US04 As a student, I want simple explanations so that complex topics are easier to understand. High

US05 As a student, I want summaries so that I can revise topics quickly. Medium

US06 As a student, I want automatically generated quizzes so that I can practice what I have learned. High

US07 As a student, I want to see my quiz results so that I can identify areas for improvement. Medium

US08 As a student, I want a dashboard so that I can view my learning activities. Medium

US09 As a student, I want the application to work on different devices so that I can study conveniently. Medium

13. Milestones

Milestone Target Deliverable

M1 Week 2 Requirements finalized

M2 Week 4 System design completed

M3 Week 6 Authentication and database completed

M4 Week 8 AI assistant functional

M5 Week 10 Quiz and summary features completed

M6 Week 12 Complete application integration

M7 Week 14 Testing completed

M8 Week 16 Final deployment and documentation

14. Team Responsibilities

Role	Responsibilities
Project Leader	Planning, coordination and progress monitoring
Frontend Developer	UI design and frontend implementation
Backend Developer	APIs, authentication and application logic
AI Developer	AI integration, prompts and AI-related functionality
Database Developer	Database design and data management
Tester	Test cases, bug identification and verification
Documentation Member	Reports, diagrams, screenshots and presentation

One student may perform multiple roles depending on the project team size.

15. Risk Management

Risk	Probability	Impact	Mitigation
AI API failure	Medium	High	Implement error handling and fallback messages
Incorrect AI responses	Medium	High	Provide appropriate prompts and validation
Database failure	Low	High	Maintain backups and proper database configuration
Schedule delays	Medium	Medium	Monitor sprint progress regularly
Integration problems	Medium	Medium	Perform integration testing after each module
Security issues	Low	High	Apply authentication and secure data-handling practices
Internet dependency	Medium	Medium	Display appropriate connection/error messages

16. Testing Plan

Testing Type	Purpose
Unit Testing	Test individual functions and components
Integration Testing	Verify communication between modules
System Testing	Test the complete application
UI Testing	Verify interface functionality and usability

Performance Testing	Check application response and behavior
Security Testing	Identify common authentication and access issues
User Acceptance Testing	Collect feedback from intended users

17. Project Deliverables

The following deliverables are planned for the project:

Project proposal.

Requirement specification document.

System architecture diagram.

Database design.

UI/UX prototype.

Source code.

AI integration module.

Test cases and test report.

Working application.

Project documentation.

Final presentation.

Project demonstration.

18. Expected Outcome

At the completion of the project, EduGenie is expected to provide students with an interactive AI-powered learning environment. Users will be able to ask questions, receive explanations, generate summaries and practice through quizzes.

The project will demonstrate how Generative AI can be integrated with conventional software components to create an educational application focused on personalized learning assistance.

19. Project Schedule Summary

Phase	Weeks	Status
Project Initiation	1	Planned
Requirement Analysis	1–2	Planned
System Design	3–4	Planned
Database & Authentication	5–6	Planned
AI Integration	7–8	Planned
Learning Features	9–10	Planned
Dashboard Integration	11–12	Planned
Testing	13–14	Planned
Deployment	15	Planned
Final Documentation	16	Planned

20. Conclusion

The Project Planning Phase establishes the foundation for the successful development of EduGenie. The project objectives, scope, requirements, modules, responsibilities, sprint schedule, milestones, risks, and deliverables have been identified.

Following the proposed Agile plan will allow the team to develop EduGenie incrementally, test each major feature, identify issues early, and produce a functional AI-powered learning assistant suitable for academic use.

Document Approval

Role	Name	Signature	Date
Project Guide	_____	_____	_____
Project Leader	_____	_____	_____
Team Member	_____	_____	_____
HOD	_____	_____	_____